



CS 304: Formal Languages and Automata Theory

Course Code	CS 304
Course Name	Formal Languages and Automata Theory
L-T-P-C	3-0-0-3
Prerequisites	CS 202, CS 208 or instructors consent
Intended for	B.Tech.
Distribution	Discipline elective for CS; CS elective for EE, ME, and Civil
Semester	5th
One-time approval	10th Senate (Jun 2016)

Course Contents

- **Regular languages** DFA, NFA, Subset construction, Regular, Pumping Lemma, DFA state minimization, Myhill-Nerode relations and theorem. [12 Lectures]
- **Grammars** Production systems, Right linear grammar and Finite state automata, Context free grammars, Normal forms, Pumping Lemma for CFLs, Subfamilies of CFL, Derivation trees and ambiguity. [10 Lectures]
- **Pushdown automata** Acceptance by final state and empty stack, Equivalence between push- down automata and context-free grammars, Closure properties of CFL, Deterministic push- down automata, the CKY algorithm. [10 Lectures]
- Advanced topics: the Chomsky-Schutzenberger theorem, Parikh's theorem.
- **Turing machines** Techniques for Turing machine construction, Generalized and restricted versions equivalent to the basic model, Universal Turing machine, Recursively enumerable sets and recursive sets. [10 Lectures]
- **Decidability** Decidable and undecidable problems, Reduction, Post's correspondence problem, Rice's theorem, decidability of membership, emptiness and equivalence problems of languages. [10 Lectures]
- **Advanced topics** Godel's incompleteness theorem, Godel's proof.

References

1. D. C. Kozen, **Automata and Computability**, Springer, 1997.
2. J. E. Hopcroft, R. Motwani and J. D. Ullman, **Introduction to automata theory, languages and computation**, 3rd Edition, Pearson, 2006.
3. E. A. Rich, Automata, **Computability and Complexity: Theory and Applications**, Pearson, 2007.
4. M. Sipser, **Introduction to the Theory of Computation**, 3rd Edition, Cengage Learning, 2012.
5. Peter Linz, **An introduction to formal language and automata**, 3rd Edition, Narosa publishing house, 2002.